A Youth Tobacco Policy Evaluation

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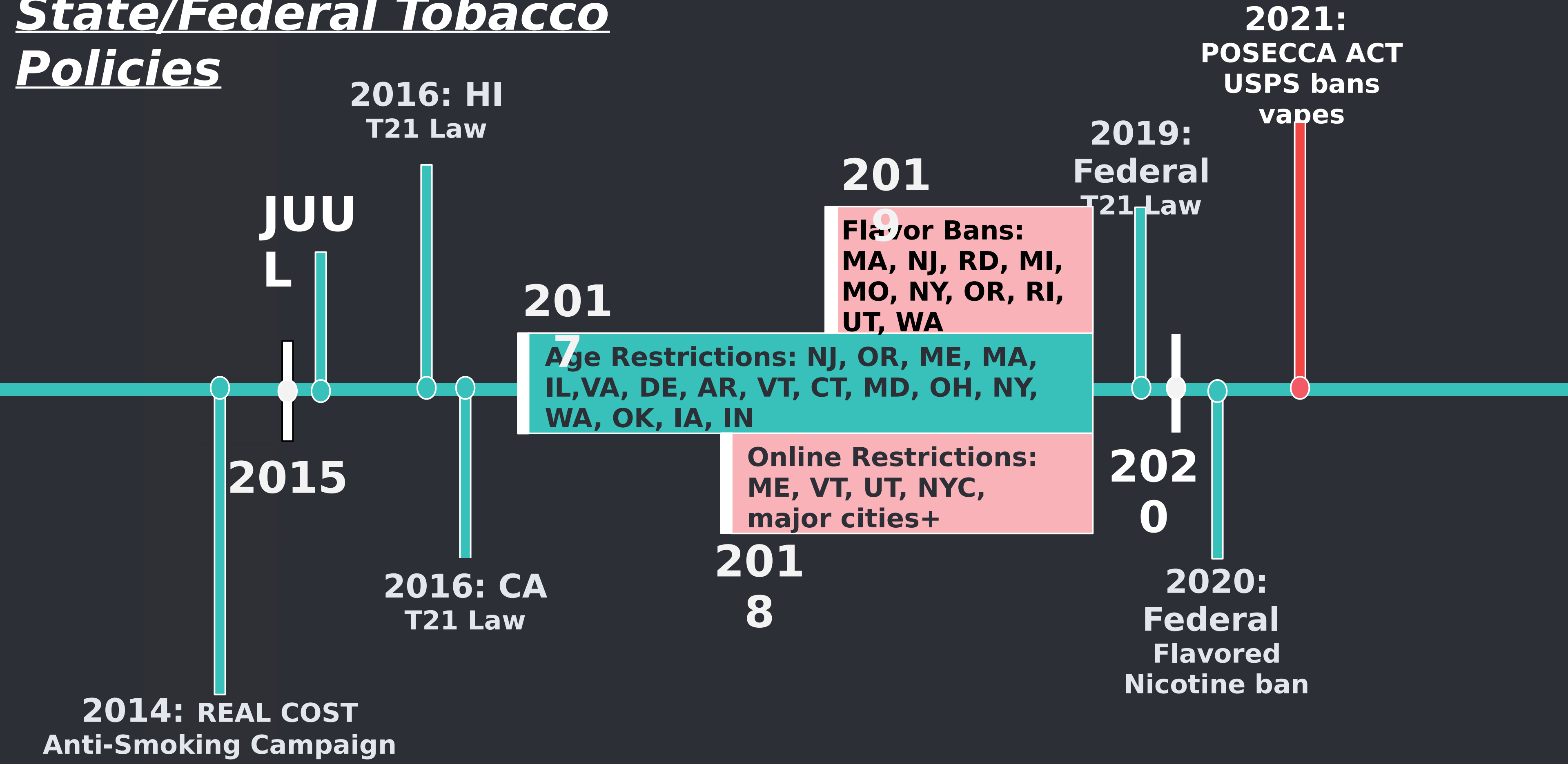
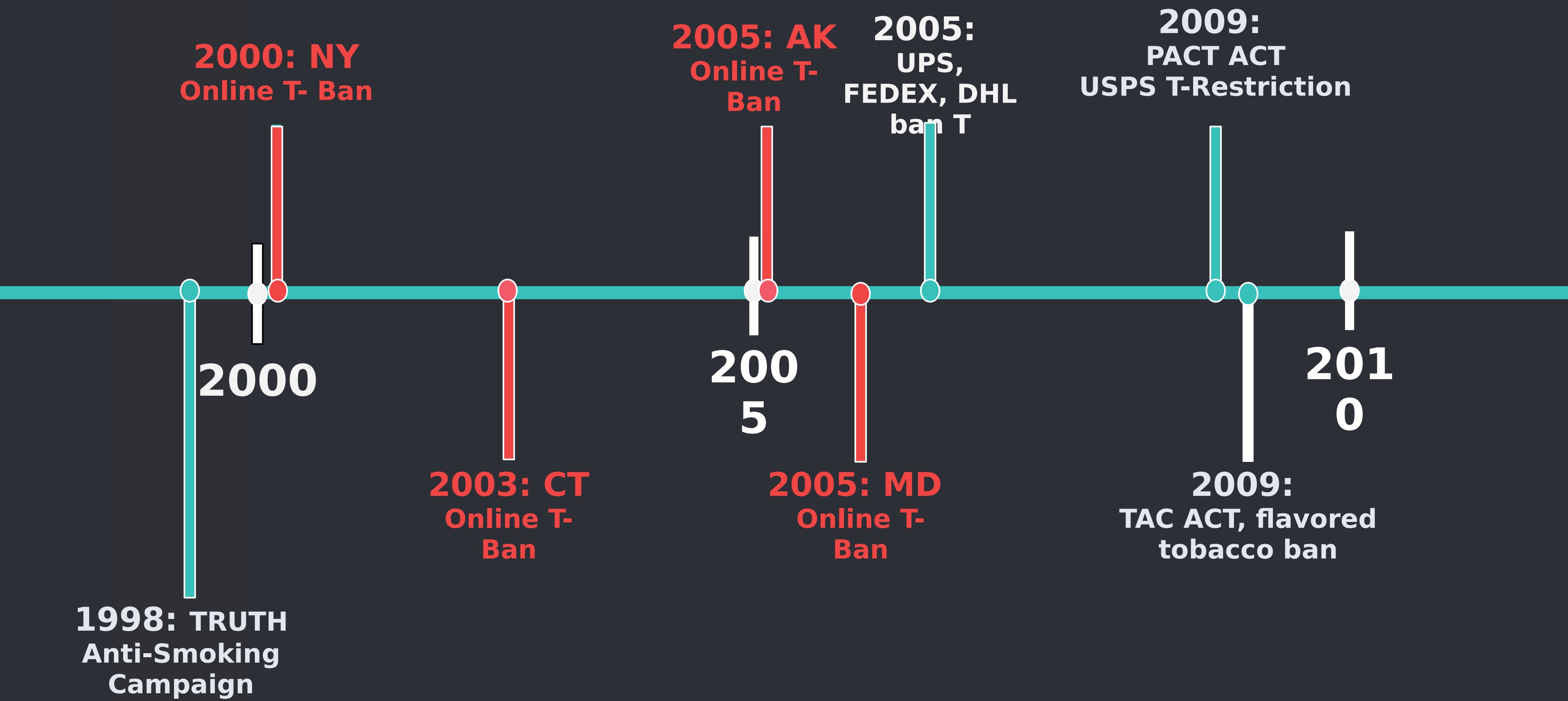
Presentation Format

- Background
- Data
- Methodology
- Results & Discussion

State/Federal Tobacco Policies



State/Federal Tobacco Policies



Why do we care about online-tobacco bans?

Background:

2009: Federally Prevent All Cigarette Trafficking (PACT) Act ***regulated*** USPS mailed tobacco

2021 Preventing Online Sales of E-Cigarettes to Children Act (POSECCA) extends ***regulations*** to vapes

- GoPuff and new services have vape delivery options

What is the best approach?

- Argument for: reduces youth nicotine use
- Argument against: legal age consumers trying to quit may not have access

Are these regulations effective?

Williams et al. (2015):

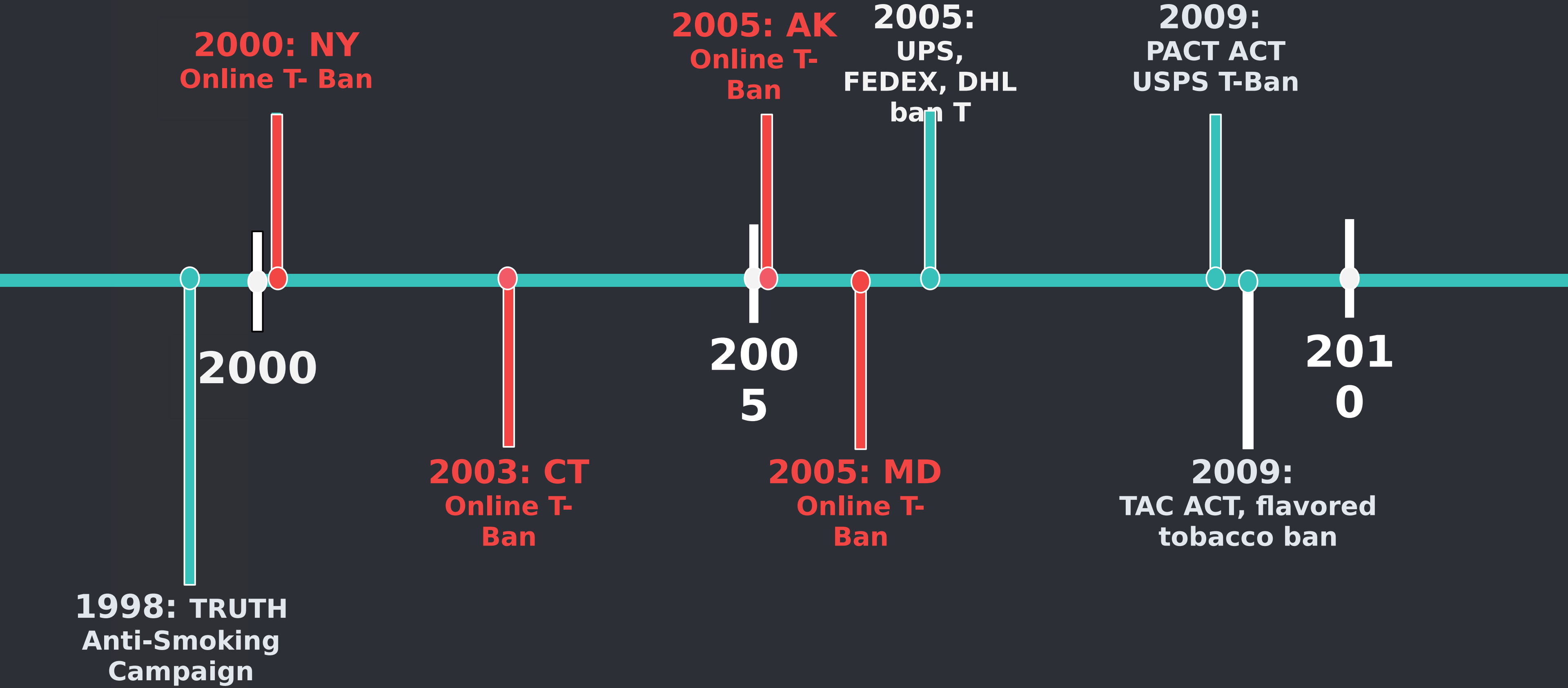
Controlled lab experiment using youth to order online tobacco

"Minors successfully received deliveries of e-cigarettes from ***76.5%*** of purchase attempts"

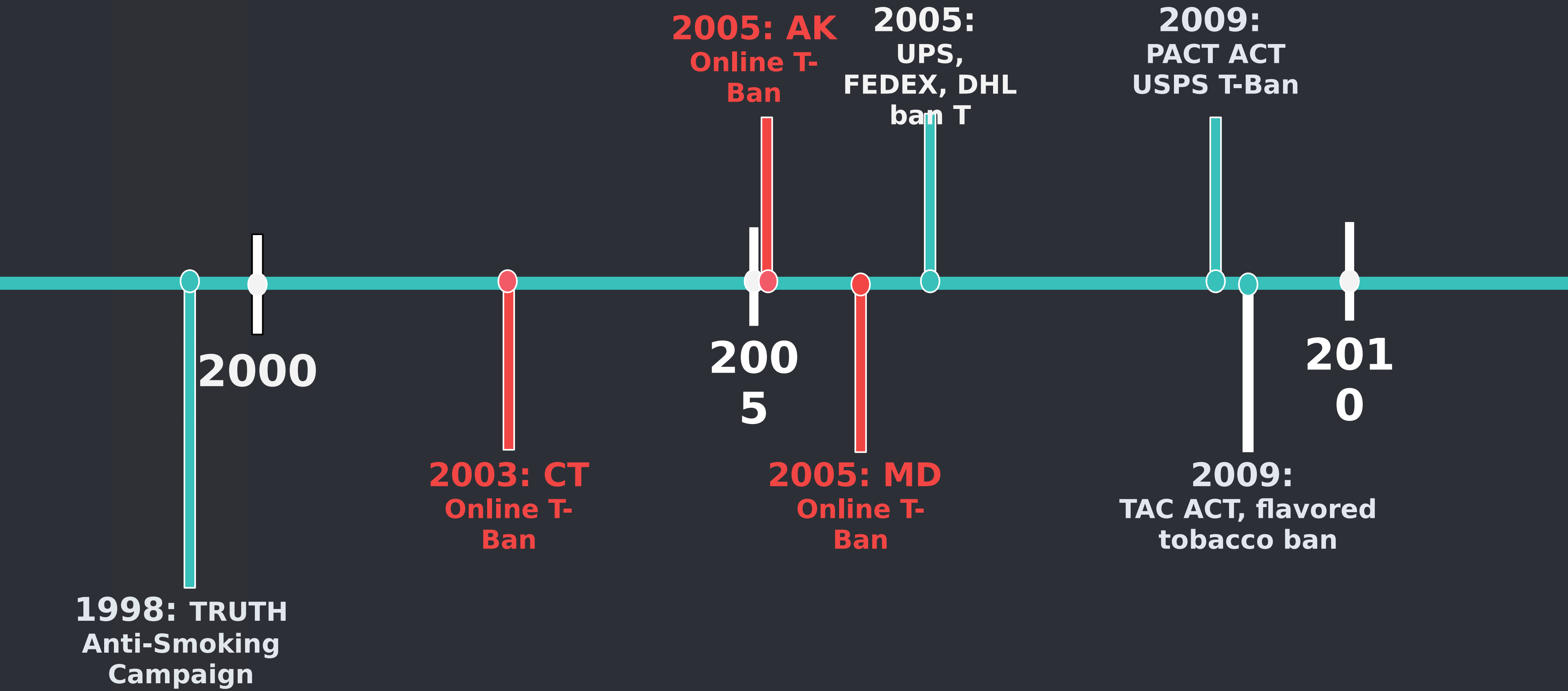
2017-2020 Policies:

"the public health community has a new reason to address illegal online sales of tobacco to youth … given the rapidly increasing use of e-cigarettes among young people" (Hamline Law)

State/Federal Tobacco Policies



State/Federal Tobacco Policies



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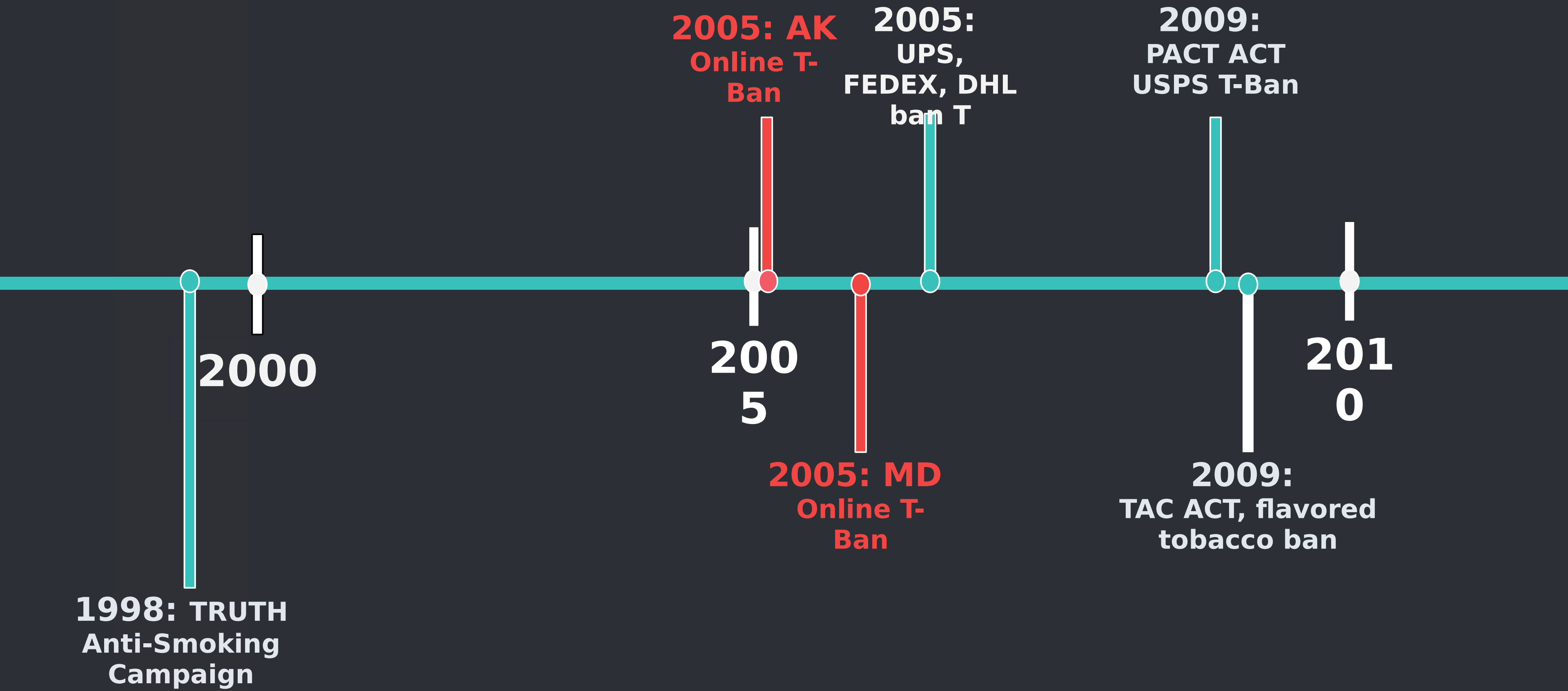

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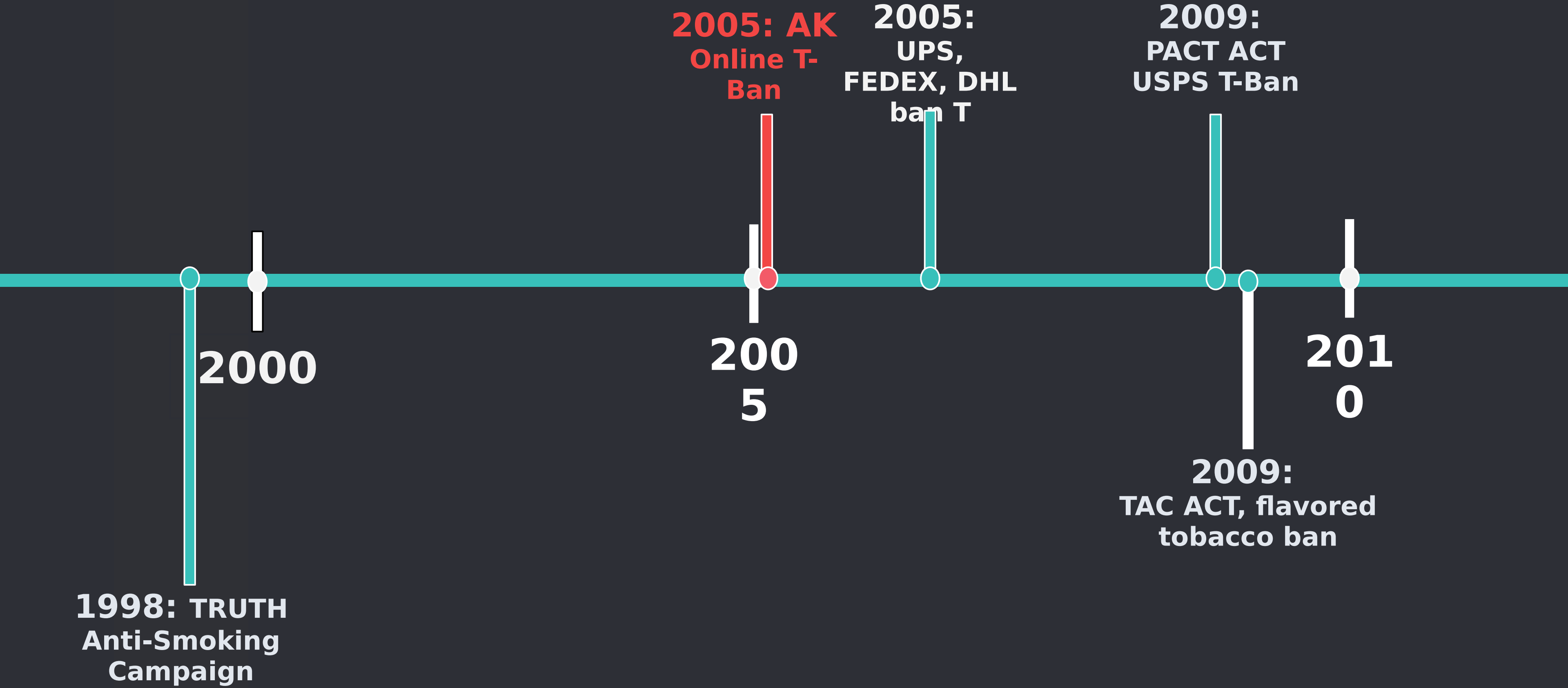

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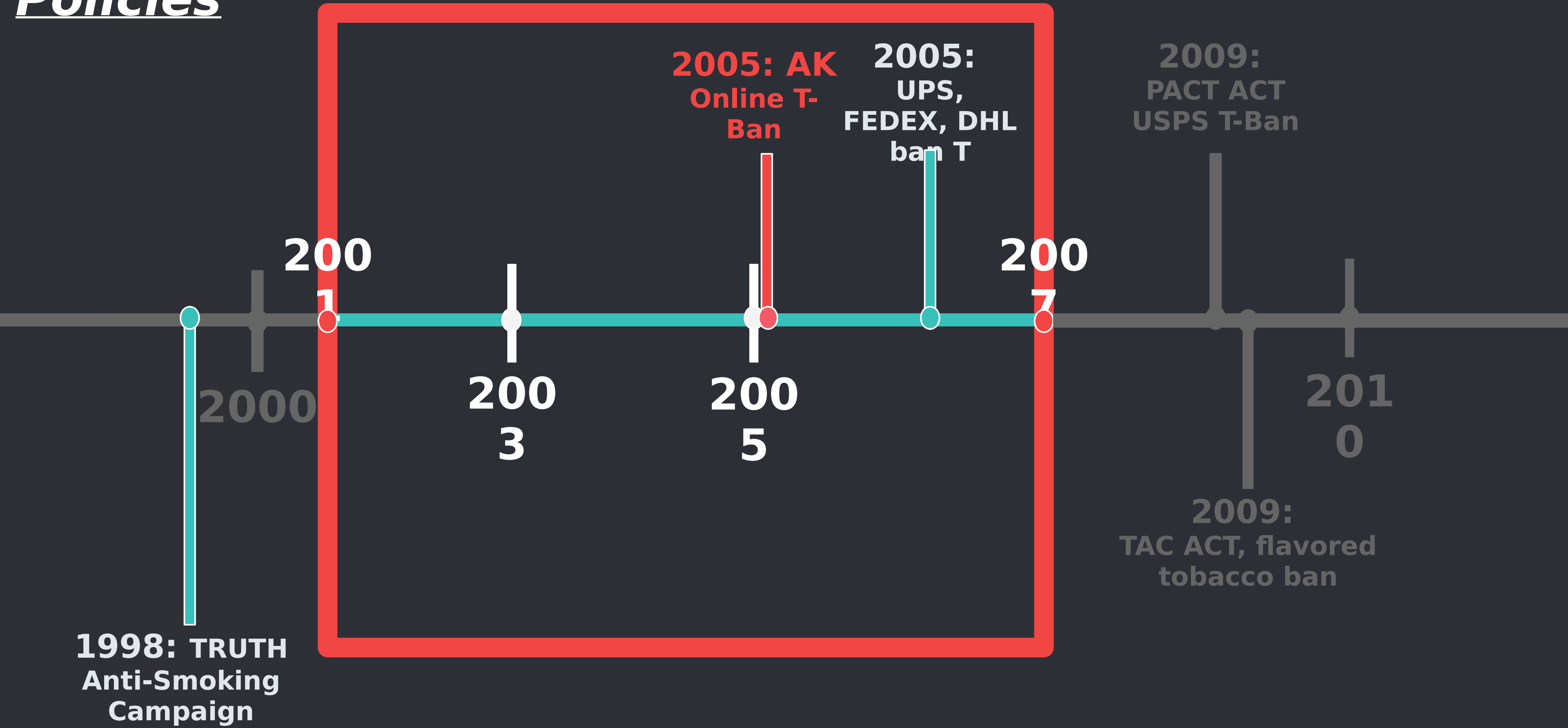

RESEARCH QUESTION:

What was the effect of the 2005 Arkansas *online ban of direct-to-consumer sales of tobacco products on youth nicotine use?*

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Data Sources

YRBS and TUS-CPS

YRBSS | Youth Risk Behavior Surveillance System

Information:

Data for ***youth*** smoking

Survey completed every 2 years by CDC in the Spring

Cross sectional data at state-level for 44 states

High school students ages <= 18

Limitations: State-level data → clustering

YRBS Data and Variables

Outcome Variables:

Cig_Any: indicator 1 if use any cigarettes in last 30 days, 0 else

Cig_Daily: indicator 1 if use cigarette daily, 0 else

Tobac_Any: indicator 1 if use tobacco products daily, 0 else

Tobac_Daily: indicator 1 if use any tobacco products, 0 else

Smoker_Any: indicator 1 if smoke any cigs per day, 0 else

Heavy_Smoker: indicator 1 if smoke 10+ cigs per day, 0 else

Demographic Variables:

Age, sex, grade

Limitation: no income variables

TUS-CPS: The Tobacco Use Supplement to the Current Population Survey

Information:

Data for ***adult*** smoking

Survey every 2-3 years by National Cancer Institute

Cross sectional data at county and state-level for all 50 states

Non-high school students ages>=18

Variables:

Outcome: Harmonized variables to match YRBS

Controls: age, grade, race, city, income

Data Continued:

Following Friedman (2021) who studied San Francisco flavored nicotine ban on youth consumption, analysis is restricted to students younger than 18 years who had ***nonmissing*** data

I repeat the same analysis with adult data using TUS-CPS

YRBS: Clustered bootstrap method at state level → limitation Bertrand et al. (2004)

- Control Group One: 38 Clusters
- Control Group Two: 8 clusters
- Control Group Three: 10 clusters
- ***Use the clustered bootstrap method***

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Methodology

Diff-in-Diff, Control Groups

DD Methodology:

EQUATION 1:

$outcome_{ist} = B_0 + B_1(ARK_{ist}) +$ $+\lambda_1 Year_t +$

$\lambda_2 States + \varepsilon_{ist}$

Where:

ARK_{ist} indicates 1 if Arkansas, 0 else

$POST_t$ indicates 1 if post policy 2005, 0 else

X_{it} are controls such as age, sex, grade

$Year_t$ state dummies for year

$outcome_{ist}$ are outcome variables such as Cig_Any and Tobac_Any

DD Methodology: Limitations

Concern:

Only one state in the sample is treated

Difficult to argue the results are not driven by Arkansas-specific trend

Methodology in other research:

Ali et al. (2019) studied effects of raising tobacco age to 21 law in Hawaii and California with only two treated states

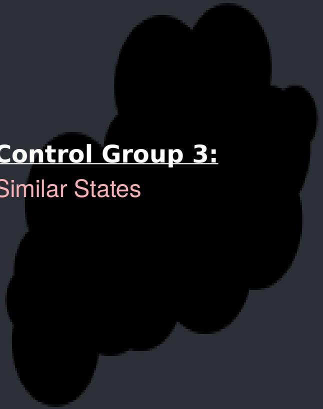
Limitation:

Is something better than nothing?

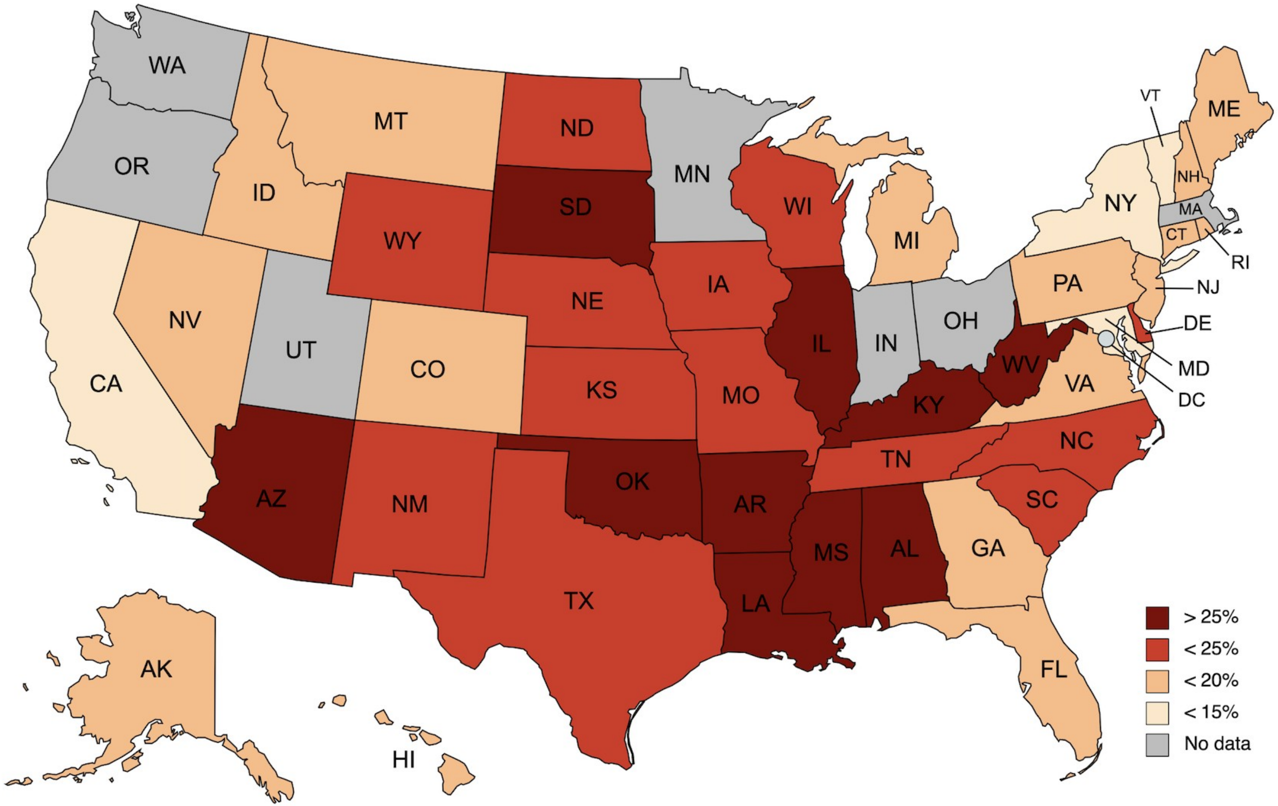
DD Methodology:

Control Groups:

Finding DD estimates robust across relevant control groups may provide evidence “parallel trends” is satisfied (Bullinger et al. (2019), Lenhart (2020)



YRBS DATA: Youth Smoking Rates 2003-2005 by State



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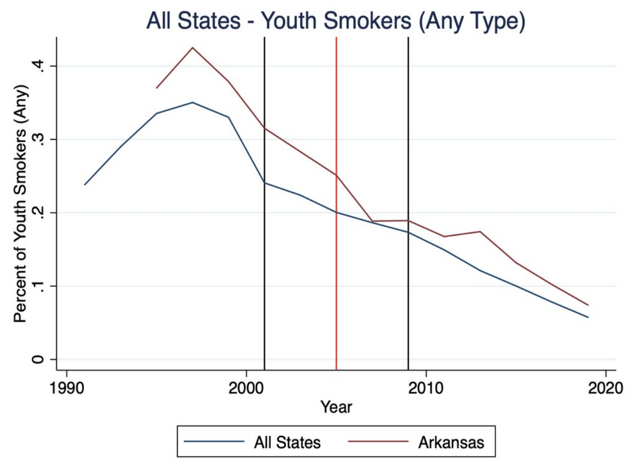
Identifying Assumptions

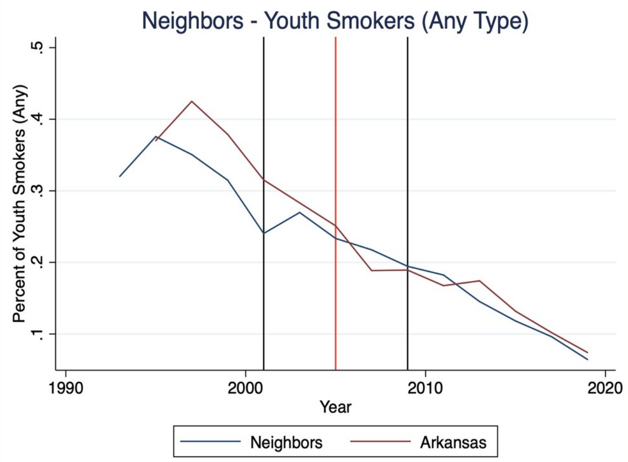
Parallel Trends

Youth Smoking Frequencies by Control Group

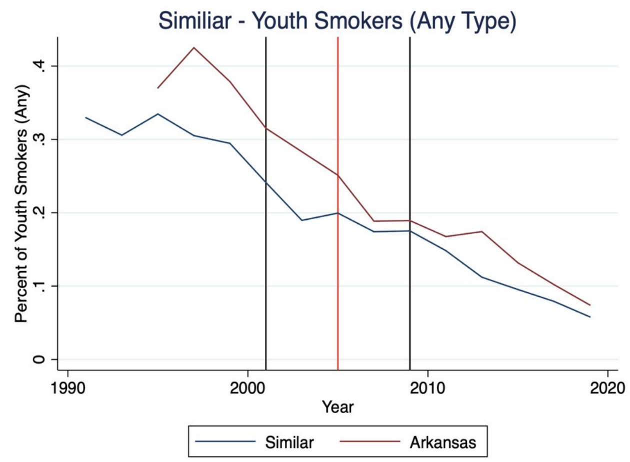
Control 1:



Control 2:

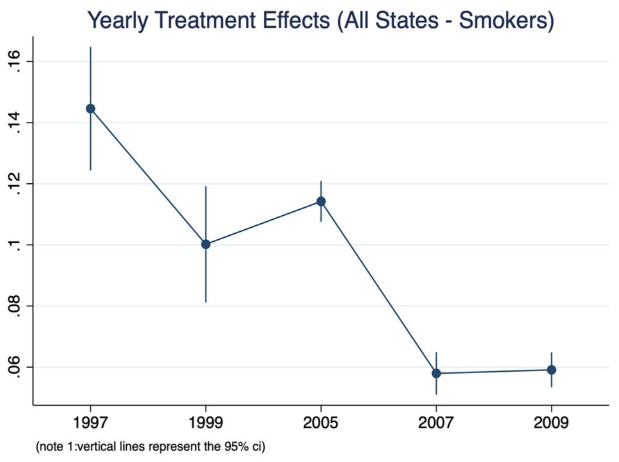


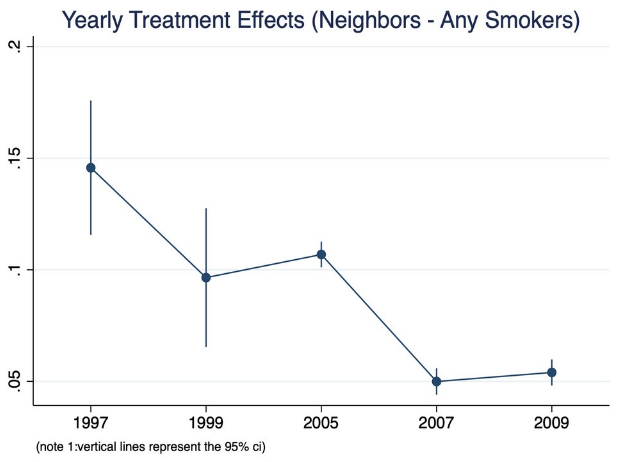
Control 3:

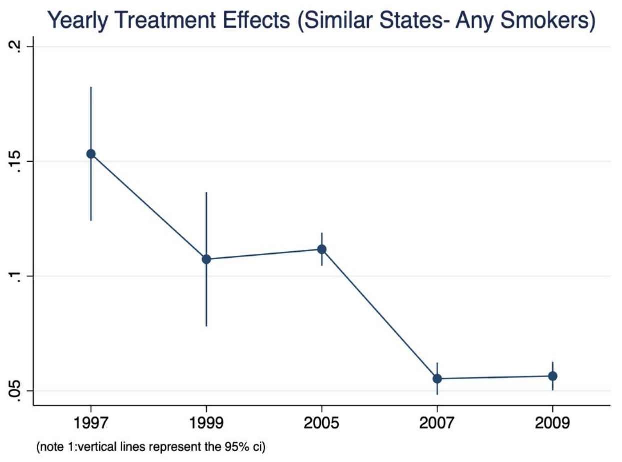


Event Study: Any Cigarette Use by Control Group

ntrol 1: **Control 2:** **Control 3:**







Evidence of parallel trends?

Satisfied?

- No other state/federal tobacco policies during time period affecting AR in time period
- Observed years pre-policy provide some evidence of little to no differential trends
- Difficult to say, some evidence but need more robustness checks

But...

- Not all states had data
- Study has one treatment state
- Error bars are larger in earlier years 1997-2001 data

Other Assumptions for this model

Key Assumption:

UPS, FEDEX, DHL ban on tobacco products affected all states equally

USPS option was still available:

> "to provide mail services to everyone, regardless of where they live, and for at least one mail product, at a uniform price."

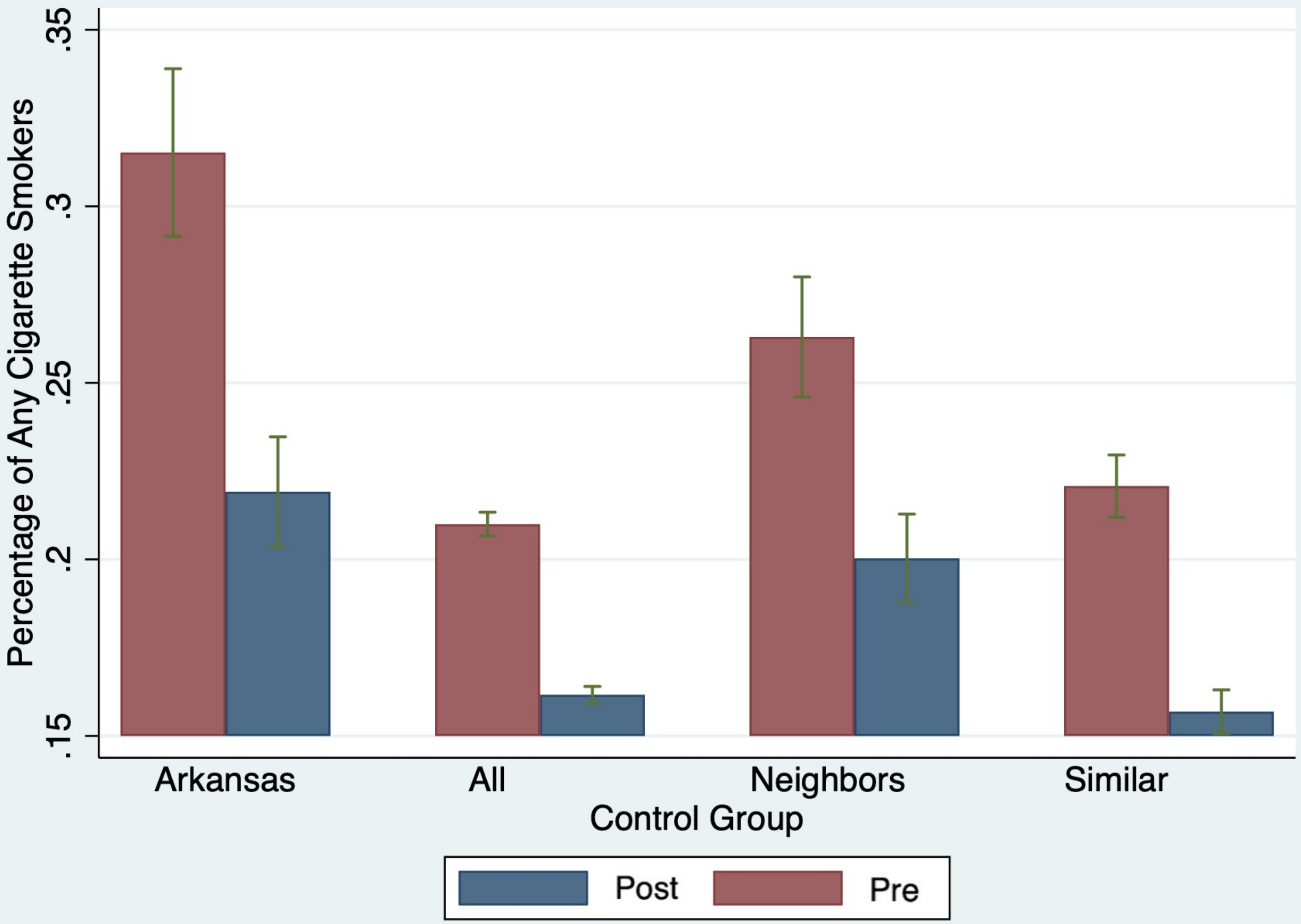
Key Assumption:

1998: TRUTH Campaign affected states similarly and not more effectatious in certain states

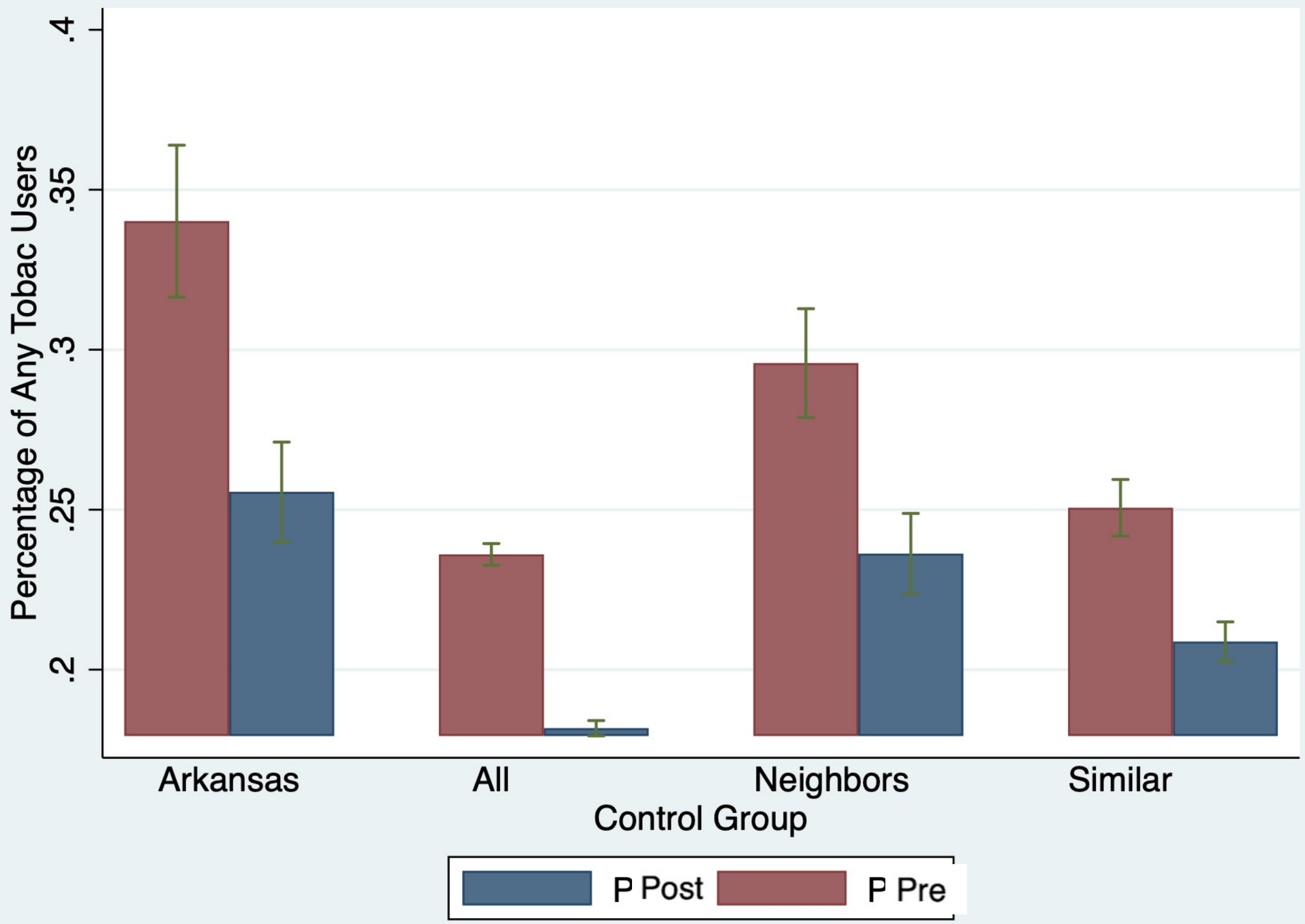
I try to control for baseline smoking rates in similar smokers in control Group 2 and 3

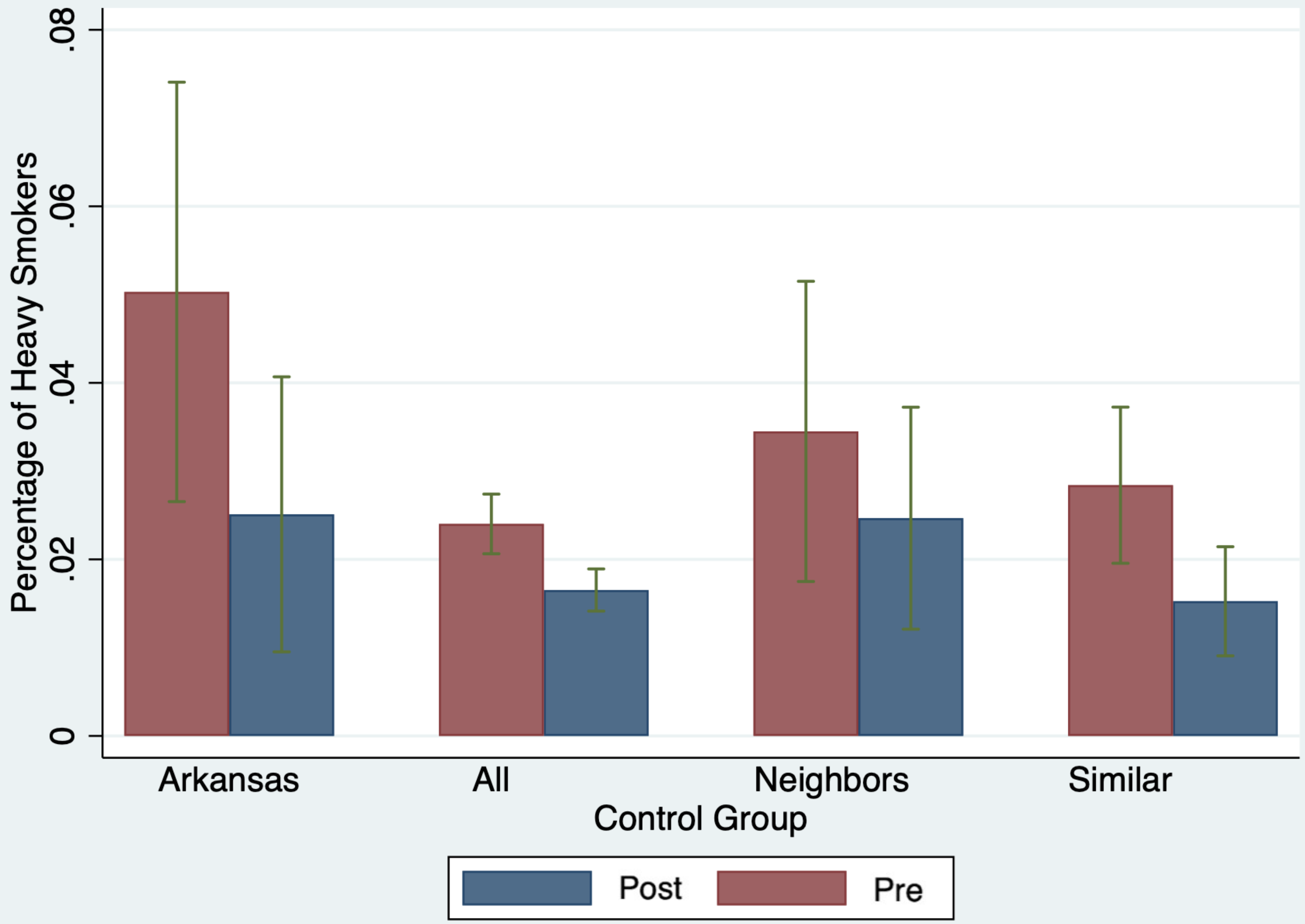
TABLE 2: Summary Stats for Youth Participants in YRBS (2001-2007)

	Treatment Group (Arkansas)	Control Group 1 (All States)	Control Group 2 (Neighboring States)	Control Group 3 (Similar States)
Aged (excludes < 12)	15.74 (0.962)	15.69 (1.010)	15.73 (0.987)	15.65 (1.027)
Male	0.474 (0.499)	0.483 (0.500)	0.476 (0.499)	0.480 (0.500)
White	0.623 (0.485)	0.613 (0.487)	0.5867(0.496)	0.484 (0.497)
Grade 9	0.344 (0.475)	0.323 (0.468)	0.344 (0.475)	0.312 (0.463)
Grade 10	0.335 (0.472)	0.303 (0.460)	0.306 (0.461)	0.302 (0.459)
Grade 11	0.254 (0.435)	0.260 (0.439)	0.255 (0.436)	0.261 (0.439)
Grade 12	0.0568 (0.231)	0.104 (0.305)	0.0887 (0.284)	0.119 (0.324)

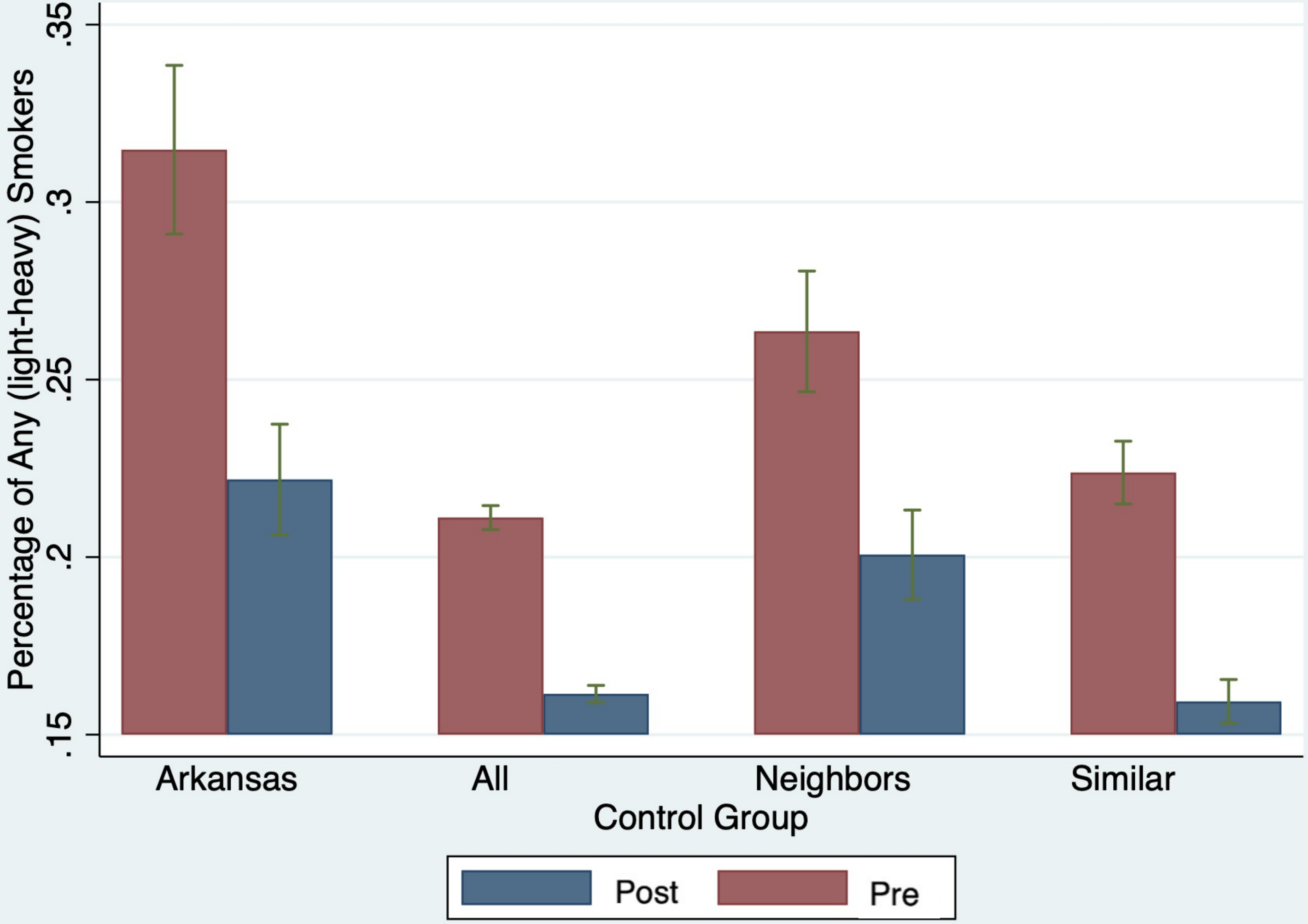


Percentage of Any Cigarette Smokers
.35
.3
.25
.2
.15
Arkansas
All
Neighbors
Similar
Control Group
Post
Pre


Percentage of Any Tobac Users
.4
.35
.3
.25
.2
Arkansas
All
Neighbors
Similar
Control Group
P Post
P Pre



Percentage of Heavy Smokers
0
.02
.04
.06
.08
Arkansas
All
Neighbors
Similar
Control Group
Post
Pre



Percentage of Any (light-heavy) Smokers
.15
.2
.25
.3
.35
Arkansas
All
Neighbors
Similar
Control Group
Post
Pre

TABLE 4: Percentage of Youth Using Cigarettes

	Control Group 1 (All States)		Control Group 2 (Neighboring States)		Control Group 3 (Similar States)	
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Daily (30 days)						
DD Estimate	-0.0353***	-0.0389***	-0.0330***	-0.0383***	-0.0323***	-0.0363***
	(0.00281)	(0.00248)	(0.00381)	(0.00372)	(0.00779)	(0.00728)
Panel B: Any (1+ day)						
DD Estimate	-0.0655***	-0.0714***	-0.0564***	-0.0632***	-0.0524***	-0.0589***
	(0.00544)	(0.00520)	(0.00677)	(0.00744)	(0.0104)	(0.00997)
Demographics		x		x		x
Year FE	x	x	x	x	x	x
State FE	x	x	x	x	x	x
Observations	230,097	182,234	42,504	42,404	52,016	51,906

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Any: Cig Use 1+ day per month

TABLE 4: Percentage of Youth Using Cigarettes

	Control Group 1 (All States)		Control Group 2 (Neighboring States)		Control Group 3 (Similar States)	
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Panel B: Any (1+ day)						
DD Estimate	-0.0655***	-0.0714***	-0.0564***	-0.0632***	-0.0524***	-0.0589***
	(0.00544)	(0.00520)	(0.00677)	(0.00744)	(0.0104)	(0.00997)
Demographics		x		x		x
Year FE	x	x	x	x	x	x
State FE	x	x	x	x	x	x
Observations	230,097	182,234	42,504	42,404	52,016	51,906

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

TABLE 5: Percentage of Youth Using Tobacco Products

	Control Group 1 (All States)		Control Group 2 (Neighboring States)		Control Group 3 (Similar States)	
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Daily (30 days)						
DD Estimate	-0.0387***	-0.0421***	-0.0343***	-0.0397***	-0.0389***	-0.0431***
	(0.00278)	(0.00236)	(0.00283)	(0.00292)	(0.00177)	(0.00155)
Panel B: Any (1+ day)						
DD Estimate	$\boxed{-0.0577^{***}}$	$\boxed{-0.0623^{***}}$	$\boxed{-0.0510^{***}}$	$\boxed{-0.0568^{***}}$	$\boxed{-0.0438^{***}}$	$\boxed{-0.0425^{***}}$
	(0.00467)	(0.00462)	(0.00500)	(0.00510)	(0.00598)	(0.00582)
Demographics		x		x		x
Year FE	x	x	x	x	x	x
State FE	x	x	x	x	x	x
Observations	182,234	181,741	42,193	42,099	42,694	42,604

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

<u>Any:</u> Tobacco Use 1+ day per month

TABLE 6: Percentage of Heavy Smoker Youth

	Control Group 1 (All States)		Control Group 2 (Neighboring States)		Control Group 3 (Similar States)	
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Heavy Smoker						
DD Estimate	-0.0262***	-0.0280***	-0.0260***	-0.0284***	-0.0265***	-0.0284***
	(0.00113)	(0.000931)	(0.00179)	(0.00135)	(0.00290)	(0.00258)
Panel B: Smoke Any						
DD Estimate	-0.0596***	-0.0657***	-0.0523***	-0.0596***	-0.0467***	-0.0535***
	(0.00597)	(0.00560)	(0.00725)	(0.00775)	(0.0107)	(0.0102)
Demographics		x		x		x
Year FE	x	x	x	x	x	x
State FE	x	x	x	x	x	x
Observations	23,969	23,927	42,515	42,418	47,765	47,659

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Heavy smoker: 10+ cigs per day

ADULTS vs Children

TABLE 7: Test - Adult (18+) Tobacco Consumption

	Control Group 1 (AllStates)		Control Group 2 (Neighboring States)		Control Group 3 (SimilarStates)	
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Adult - Any Cig (1+ day)						
DD Estimate	0.00209	0.00935*	-0.0202***	-0.0181***	-0.000505	0.00954
	(0.00390)	(0.00521)	(0.00231)	(0.00235)	(0.00779)	(0.00988)
	12,656	12,656	1,897	1,897	3,170	3,170
Panel B: Youth - Any Cig (1+ day)						
DD Estimate	-0.0655***	-0.0714***	-0.0564***	-0.0632***	-0.0524***	-0.0589***
	(0.00544)	(0.00520)	(0.00677)	(0.00744)	(0.0104)	(0.00997)
	230,767	230,097	42,504	42,404	52,016	51,906

Any Cig: indicator = 1 if use any cigarettes in last 30 days, 0 else

ADULTS vs Children

Panel C: Adult Smoker Any						
DD Estimate	-0.0280***	-0.0338***	-0.0385***	-0.0339***	-0.0197***	-0.0294***
	(0.00235)	(0.00196)	(0.00580)	(0.00325)	(0.00508)	(0.00484)
	156,856	156,856	22,683	22,683	34,884	34,884
Panel D: Youth Smoker Any						
DD Estimate	-0.0596***	-0.0657***	-0.0523***	-0.0596***	-0.0467***	-0.0535***
	(0.00597)	(0.00560)	(0.00725)	(0.00775)	(0.0107)	(0.0102)
	213,440	212,796	42,515	42,418	47,765	47,659
Demographics		x		x		x
Year FE	x	x	x	x	x	x
State FE	x	x	x	x	x	x

*** p<0.01, ** p<0.05, * p<0.1

Smoke Any: indicator = 1 if smoke 1+ cigs per day, 0 else

5-6 percentage point decrease in the percent of students that smoke anything, controlling for age, sex, and education.

2.5-2.8 percentage point decrease in the percent of students that are heavy smokers, controlling for age, sex, and education.

Non-frequent adults smokers are less affected than infrequent youth smokers

What have we learned?

<u>Takeaways</u>

- Some evidence that this change may have been driven by the policy change
  - Adults seemingly affected less than youth
  - Some evidence in parallel trends
  - Control group estimates similar
- *If results were robust, they may be a lower bound estimate for a policy today*

<u>Limitations</u>:

- It is still difficult to ensure that the effect was not due to any within state changes in AR
- Efficacy of anti-smoking campaign may impact AR differently than other states because of the high baseline

Thanks!